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| 25 | <p>In Fig. 25.1 below, a beam of particles, each of charge q and mass m, is travelling at speed v through a region in which a magnetic field B is perpendicular to an electric field E. The beam of particles travels undeflected in this region.</p> <p>In Fig. 25.2 below, the electric field is switched off. The beam is found to form an arc of a circle of radius r. The effects of gravity is negligible.</p> |
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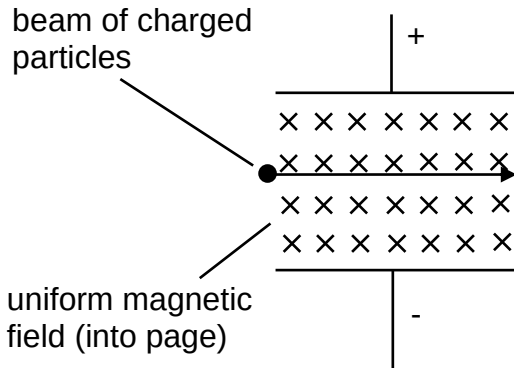


Fig. 25.1

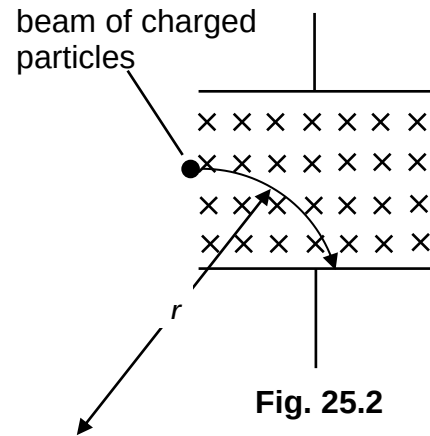


Fig. 25.2

Derive an expression for the mass m of the particle.

A

$$\frac{Bqr}{E}$$

B

$$\frac{Eqr}{v}$$

C

$$\frac{B^2qr}{E}$$

D

$$\frac{B^2q}{Er}$$

Answer: C

When the charged particles enter the cross field, it passes through undeflected. Hence,

$$F_E = F_B$$

$$qE = Bqv \quad \text{----- (1)}$$

$$v = \frac{E}{B}$$

When the electric field is switched off, the magnetic force provides the centripetal force.

$$Bqv = \frac{mv^2}{r} \quad \text{----- (2)}$$

Sub (1) into (2)

$$Bqv = \frac{mv^2}{r}$$

$$m = \frac{Bqr}{v}$$

$$= \frac{Bqr}{\left(\frac{E}{B}\right)}$$

$$= \frac{B^2qr}{E}$$